

CS410 Midterm 2 Part B

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I. QUESTION (10 POINTS)

Give context-free grammar that generate the following language. The alphabet is $\{0, 1\}$.

$A = \{w \mid w \text{ contains at least two 0s and at most two 1s}\}$

II. QUESTION (15 POINTS)

Give PDA that generate the following language. The alphabet is $\{0, 1\}$.

$B = \{w \mid w \text{ contains at least two 0s and at most two 1s}\}$

III. QUESTION (15 POINTS)

Give a CFG generating the language of strings with twice as many a's as b's. The alphabet is $\{a, b\}$.

IV. QUESTION (10 POINTS)

Give a PDA generating the language of strings with twice as many a's as b's. The alphabet is $\{a, b\}$.